Appendix

Group 3-

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Sources.csv table

file	row_i	column	value	source_ty	source_or_calculation	sanity_check
products.csv	PH-FLAG	product_name	Samsung Galaxy S25 Ultra	public	Samsung product announcement identifies the Galaxy S25 Ultra; the 256 GB storage variant is the project product definition.	Product name and storage variant are consistent with the product record used by the simulation.
products.csv	PH-FLAG	frame_material	Titanium	public	Samsung product specification states a titanium frame; official product page is recorded in this	Material label is consistent with the selected Galaxy S25 Ultra product specification.
products.csv	PH-FLAG	mass_kg	0.218	public	Samsung specification: 218 g. Unit conversion: 218 g ÷ 1,000 = 0.218 kg.	Conversion is dimensionally correct; 0.218 kg converts back to 218 g and matches products.csv.
products.csv	PH-FLAG	list_price_eur	755.46	guesstimate	MediaMarkt Germany retail price: EUR 899.00 including 19% VAT. Net-price calculation: EUR 899.00 ÷ 1.19 = EUR 755.46. This is a dated retail-	Arithmetic check: EUR 755.46 × 1.19 = approximately EUR 899.00. Treat as a price snapshot because retail prices can change.
workcenters.cs	WC1	num_servers	1	guesstimate	Model assumption: 1 parallel server(s)/machine(s) assigned to the SMT workcenter; no plant-specific	Capacity check uses 1 server(s) in the workcenter model; simulated utilization for this station is 83%,
workcenters.cs	WC1	shift_hours	8	guesstimate	Model assumption: one 8-hour shift per operating day; selected to define the weekly production	8 hours/day × 5 days/week = 40 scheduled hours/week, consistent with the simulation
workcenters.cs	WC1	days_per_week	5	guesstimate	Model assumption: the production line operates five days per week (Monday–Friday).	5 operating days/week × 8 hours/day = 40 scheduled hours/week.
workcenters.cs	WC1	availability	0.9	guesstimate	Model assumption: 90% technical availability; the remaining 10% represents planned/unplanned	Availability is between 0 and 1; effective weekly capacity is 40 h × 0.90 = 36.0 h per server before
workcenters.cs	WC1	process_time_cv	0.5	guesstimate	Model assumption: coefficient of variation (CV) of process time = 0.5; represents relative process-	CV is non-negative and dimensionless; the selected value is a simulation assumption, not an
workcenters.cs	WC1	fixed_cost_eur_w	2500	guesstimate	Model assumption: weekly fixed operating cost of EUR 2,500.00 assigned to SMT; used in the	Cost is positive and expressed per week; included as a fixed workcenter cost rather than a per-unit
workcenters.cs	WC2	num_servers	1	guesstimate	Model assumption: 1 parallel server(s)/machine(s) assigned to the Board test workcenter; no plant-	Capacity check uses 1 server(s) in the workcenter model; simulated utilization for this station is 37%,
workcenters.cs	WC2	shift_hours	8	guesstimate	Model assumption: one 8-hour shift per operating day; selected to define the weekly production	8 hours/day × 5 days/week = 40 scheduled hours/week, consistent with the simulation
workcenters.cs	WC2	days_per_week	5	guesstimate	Model assumption: the production line operates five days per week (Monday–Friday).	5 operating days/week × 8 hours/day = 40 scheduled hours/week.
workcenters.cs	WC2	availability	0.9	guesstimate	Model assumption: 90% technical availability; the remaining 10% represents planned/unplanned	Availability is between 0 and 1; effective weekly capacity is 40 h × 0.90 = 36.0 h per server before
workcenters.csv	WC2	process_time_cv	0.3	guesstimate	Model assumption: coefficient of variation (CV) of process time = 0.3; represents relative process-time variability at Board test. CV is standard	CV is non-negative and dimensionless; the selected value is a simulation assumption, not an empirically measured plant statistic.
workcenters.cs	WC2	fixed_cost_eur_w	800	guesstimate	Model assumption: weekly fixed operating cost of EUR 800.00 assigned to Board test; used in the	Cost is positive and expressed per week; included as a fixed workcenter cost rather than a per-unit
workcenters.cs	WC3	num_servers	2	guesstimate	Model assumption: 2 parallel server(s)/machine(s) assigned to the Assembly workcenter; no plant-	Capacity check uses 2 server(s) in the workcenter model; simulated utilization for this station is 49%,
workcenters.cs	WC3	shift_hours	8	guesstimate	Model assumption: one 8-hour shift per operating day; selected to define the weekly production	8 hours/day × 5 days/week = 40 scheduled hours/week, consistent with the simulation
workcenters.cs	WC3	days_per_week	5	guesstimate	Model assumption: the production line operates five days per week (Monday–Friday).	5 operating days/week × 8 hours/day = 40 scheduled hours/week.
workcenters.cs	WC3	availability	0.85	guesstimate	Model assumption: 85% technical availability; the remaining 15% represents planned/unplanned	Availability is between 0 and 1; effective weekly capacity is 40 h × 0.85 = 34.0 h per server before
workcenters.csv	WC3	process_time_cv	0.4	guesstimate	Model assumption: coefficient of variation (CV) of process time = 0.4; represents relative process-time variability at Assembly. CV is standard	CV is non-negative and dimensionless; the selected value is a simulation assumption, not an empirically measured plant statistic.
workcenters.cs	WC3	fixed_cost_eur_w	600	guesstimate	Model assumption: weekly fixed operating cost of EUR 600.00 assigned to Assembly; used in the	Cost is positive and expressed per week; included as a fixed workcenter cost rather than a per-unit
workcenters.cs	WC4	num_servers	1	guesstimate	Model assumption: 1 parallel server(s)/machine(s) assigned to the Final test workcenter; no plant-	Capacity check uses 1 server(s) in the workcenter model; simulated utilization for this station is 29%,
workcenters.cs	WC4	shift_hours	8	guesstimate	Model assumption: one 8-hour shift per operating day; selected to define the weekly production	8 hours/day × 5 days/week = 40 scheduled hours/week, consistent with the simulation
workcenters.cs	WC4	days_per_week	5	guesstimate	Model assumption: the production line operates five days per week (Monday–Friday).	5 operating days/week × 8 hours/day = 40 scheduled hours/week.
workcenters.cs	WC4	availability	0.9	guesstimate	Model assumption: 90% technical availability; the remaining 10% represents planned/unplanned	Availability is between 0 and 1; effective weekly capacity is 40 h × 0.90 = 36.0 h per server before
workcenters.csv	WC4	process_time_cv	0.3	guesstimate	Model assumption: coefficient of variation (CV) of process time = 0.3; represents relative process-time variability at Final test. CV is standard	CV is non-negative and dimensionless; the selected value is a simulation assumption, not an empirically measured plant statistic.
workcenters.cs	WC4	fixed_cost_eur_w	1000	guesstimate	Model assumption: weekly fixed operating cost of EUR 1,000.00 assigned to Final test; used in the	Cost is positive and expressed per week; included as a fixed workcenter cost rather than a per-unit
workcenters.cs	WC5	num_servers	1	guesstimate	Model assumption: 1 parallel server(s)/machine(s) assigned to the Packaging workcenter; no plant-	Capacity check uses 1 server(s) in the workcenter model; simulated utilization for this station is 14%,
workcenters.cs	WC5	shift_hours	8	guesstimate	Model assumption: one 8-hour shift per operating day; selected to define the weekly production	8 hours/day × 5 days/week = 40 scheduled hours/week, consistent with the simulation
workcenters.cs	WC5	days_per_week	5	guesstimate	Model assumption: the production line operates five days per week (Monday–Friday).	5 operating days/week × 8 hours/day = 40 scheduled hours/week.
workcenters.cs	WC5	availability	0.95	guesstimate	Model assumption: 95% technical availability; the remaining 5% represents planned/unplanned	Availability is between 0 and 1; effective weekly capacity is 40 h × 0.95 = 38.0 h per server before

workcenters.csv	WC5	process_time_cv	0.2	guesstimate	Model assumption: coefficient of variation (CV) of process time = 0.2; represents relative process-time variability at Packaging. CV is standard	CV is non-negative and dimensionless; the selected value is a simulation assumption, not an empirically measured plant statistic.
workcenters.cs	WC5	fixed_cost_eur_w	300	guesstimate	Model assumption: weekly fixed operating cost of EUR 300.00 assigned to Packaging; used in the	Cost is positive and expressed per week; included as a fixed workcenter cost rather than a per-unit
routing.csv	PH-FLAG	process_time_mi	2	guesstimate	Model assumption: 2.0 minute(s) processing time per unit at routing sequence PH-FLAG; no	Processing time is positive and expressed in minutes per unit; it is used to calculate workcenter
routing.csv	PH-FLAG	setup_time_min	0	guesstimate	Model assumption: zero setup time per routing step because the baseline model does not	Setup time is non-negative; zero means no additional changeover time is added to the
routing.csv	PH-FLAG	process_time_mi	1	guesstimate	Model assumption: 1.0 minute(s) processing time per unit at routing sequence PH-FLAG; no	Processing time is positive and expressed in minutes per unit; it is used to calculate workcenter
routing.csv	PH-FLAG	setup_time_min	0	guesstimate	Model assumption: zero setup time per routing step because the baseline model does not	Setup time is non-negative; zero means no additional changeover time is added to the
routing.csv	PH-FLAG	process_time_mi	3	guesstimate	Model assumption: 3.0 minute(s) processing time per unit at routing sequence PH-FLAG; no	Processing time is positive and expressed in minutes per unit; it is used to calculate workcenter
routing.csv	PH-FLAG	setup_time_min	0	guesstimate	Model assumption: zero setup time per routing step because the baseline model does not	Setup time is non-negative; zero means no additional changeover time is added to the
routing.csv	PH-FLAG	process_time_mi	1.5	guesstimate	Model assumption: 1.5 minute(s) processing time per unit at routing sequence PH-FLAG; no	Processing time is positive and expressed in minutes per unit; it is used to calculate workcenter
routing.csv	PH-FLAG	setup_time_min	0	guesstimate	Model assumption: zero setup time per routing step because the baseline model does not	Setup time is non-negative; zero means no additional changeover time is added to the
routing.csv	PH-FLAG	process_time_mi	0.5	guesstimate	Model assumption: 0.5 minute(s) processing time per unit at routing sequence PH-FLAG; no	Processing time is positive and expressed in minutes per unit; it is used to calculate workcenter
routing.csv	PH-FLAG	setup_time_min	0	guesstimate	Model assumption: zero setup time per routing step because the baseline model does not	Setup time is non-negative; zero means no additional changeover time is added to the
process_environment.	WC1	kw_running	5	guesstimate	Model assumption: 5.0 kW electrical demand for workcenter WC1 in the running state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC1	kw_idle	1	guesstimate	Model assumption: 1.0 kW electrical demand for workcenter WC1 in the idle state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC1	waste_kg_per_unit	0.02	guesstimate	Model assumption: 0.02 kg process waste per unit at SMT; used as a per-unit environmental input.	Waste quantity is non-negative and expressed in kg per unit; zero indicates no separately modeled waste for this step, not proof of zero real-world
process_environment.	WC1	water_l_per_unit	0.3	guesstimate	Model assumption: 0.3 litres of process water per unit at SMT; no metered water data was supplied.	Water use is non-negative and expressed in litres per unit; zero means water use is not represented for this workcenter in the model.
process_environment.	WC2	kw_running	1	guesstimate	Model assumption: 1.0 kW electrical demand for workcenter WC2 in the running state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC2	kw_idle	0.2	guesstimate	Model assumption: 0.2 kW electrical demand for workcenter WC2 in the idle state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC2	waste_kg_per_unit	0	guesstimate	Model assumption: 0.00 kg process waste per unit at Board test; used as a per-unit environmental input.	Waste quantity is non-negative and expressed in kg per unit; zero indicates no separately modeled waste for this step, not proof of zero real-world
process_environment.	WC2	water_l_per_unit	0	guesstimate	Model assumption: 0.0 litres of process water per unit at Board test; no metered water data was supplied.	Water use is non-negative and expressed in litres per unit; zero means water use is not represented for this workcenter in the model.
process_environment.	WC3	kw_running	0.5	guesstimate	Model assumption: 0.5 kW electrical demand for workcenter WC3 in the running state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC3	kw_idle	0.1	guesstimate	Model assumption: 0.1 kW electrical demand for workcenter WC3 in the idle state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC3	waste_kg_per_unit	0.01	guesstimate	Model assumption: 0.01 kg process waste per unit at Assembly; used as a per-unit environmental input.	Waste quantity is non-negative and expressed in kg per unit; zero indicates no separately modeled waste for this step, not proof of zero real-world
process_environment.	WC3	water_l_per_unit	0	guesstimate	Model assumption: 0.0 litres of process water per unit at Assembly; no metered water data was supplied.	Water use is non-negative and expressed in litres per unit; zero means water use is not represented for this workcenter in the model.
process_environment.	WC4	kw_running	1	guesstimate	Model assumption: 1.0 kW electrical demand for workcenter WC4 in the running state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC4	kw_idle	0.2	guesstimate	Model assumption: 0.2 kW electrical demand for workcenter WC4 in the idle state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC4	waste_kg_per_unit	0	guesstimate	Model assumption: 0.00 kg process waste per unit at Final test; used as a per-unit environmental input.	Waste quantity is non-negative and expressed in kg per unit; zero indicates no separately modeled waste for this step, not proof of zero real-world

process_environment.	WC4	water_l_per_unit	0	guesstimate	Model assumption: 0.0 litres of process water per unit at Final test; no metered water data was supplied.	Water use is non-negative and expressed in litres per unit; zero means water use is not represented for this workcenter in the model.
process_environment.	WC5	kw_running	0.2	guesstimate	Model assumption: 0.2 kW electrical demand for workcenter WC5 in the running state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC5	kw_idle	0.05	guesstimate	Model assumption: 0.05 kW electrical demand for workcenter WC5 in the idle state; no machine-specific meter data was supplied.	Power demand is non-negative and expressed in kW; running demand is not lower than idle demand for this workcenter.
process_environment.	WC5	waste_kg_per_unit	0.03	guesstimate	Model assumption: 0.03 kg process waste per unit at Packaging; used as a per-unit environmental input.	Waste quantity is non-negative and expressed in kg per unit; zero indicates no separately modeled waste for this step, not proof of zero real-world
process_environment.	WC5	water_l_per_unit	0	guesstimate	Model assumption: 0.0 litres of process water per unit at Packaging; no metered water data was supplied.	Water use is non-negative and expressed in litres per unit; zero means water use is not represented for this workcenter in the model.
suppliers.csv	S1	unit_price_eur	70	public	Assumed/quoted unit price recorded for MemCore (representative South Korean memory supplier): EUR 70 per memory component; see	Positive EUR/unit value; supplier prices are compared on the same per-component basis in the material-cost calculation.
suppliers.csv	S1	promised_lead_da	15	guesstimate	Model estimate of 15 promised lead days for MemCore (representative South Korean memory	Promised lead time (15 days) is not greater than the modeled mean lead time (35 days).
suppliers.csv	S1	lead_time_mean_days	35	guesstimate	Model estimate for MemCore (representative South Korean memory supplier): 30 days sea transit on L1-SEA plus 5 days handling/scheduling	Cross-check against network_lanes.csv: route transit component is 30 days; remaining allowance is 5 days. Total matches suppliers.csv.
suppliers.csv	S1	lead_time_sd_days	4	guesstimate	Model assumption: standard deviation of supplier lead time = 4 days for MemCore (representative South Korean memory supplier); used to	Standard deviation is non-negative and smaller than the corresponding mean lead time; it is a variability assumption, not a measured
suppliers.csv	S1	capacity_per_month	60000	guesstimate	Model assumption: 60,000 memory components/month available from MemCore (representative South Korean memory supplier);	Capacity is greater than the recorded minimum order quantity (20,000/month) and is expressed in units/month.
suppliers.csv	S1	min_per_month	20000	guesstimate	Model assumption: minimum monthly order quantity of 20,000 memory components from	Minimum is positive and does not exceed the modeled monthly capacity (60,000 units/month).
suppliers.csv	S1	kg_co2e_per_unit	0.8	guesstimate	Model estimate: 0.8 kg CO2e per memory component supplied by MemCore (representative South Korean memory supplier); supplier-specific	Emission factor is non-negative and expressed per component; used only for the memory-chip supplier-emissions part of the model.
suppliers.csv	S2	unit_price_eur	72	guesstimate	Assumed/quoted unit price recorded for MemAsia (representative Taiwanese memory supplier): EUR	Positive EUR/unit value; supplier prices are compared on the same per-component basis in
suppliers.csv	S2	promised_lead_da	12	guesstimate	Model estimate of 12 promised lead days for MemAsia (representative Taiwanese memory	Promised lead time (12 days) is not greater than the modeled mean lead time (48 days).
suppliers.csv	S2	lead_time_mean_days	48	guesstimate	Model estimate for MemAsia (representative Taiwanese memory supplier): 43 days sea transit on L2-SEA plus 5 days handling/scheduling	Cross-check against network_lanes.csv: route transit component is 43 days; remaining allowance is 5 days. Total matches suppliers.csv.
suppliers.csv	S2	lead_time_sd_days	3	guesstimate	Model assumption: standard deviation of supplier lead time = 3 days for MemAsia (representative Taiwanese memory supplier); used to represent	Standard deviation is non-negative and smaller than the corresponding mean lead time; it is a variability assumption, not a measured
suppliers.csv	S2	capacity_per_month	50000	guesstimate	Model assumption: 50,000 memory components/month available from MemAsia (representative Taiwanese memory supplier); no	Capacity is greater than the recorded minimum order quantity (10,000/month) and is expressed in units/month.
suppliers.csv	S2	min_per_month	10000	guesstimate	Model assumption: minimum monthly order quantity of 10,000 memory components from	Minimum is positive and does not exceed the modeled monthly capacity (50,000 units/month).
suppliers.csv	S2	kg_co2e_per_unit	0.7	guesstimate	Model estimate: 0.7 kg CO2e per memory component supplied by MemAsia (representative Taiwanese memory supplier); supplier-specific	Emission factor is non-negative and expressed per component; used only for the memory-chip supplier-emissions part of the model.
suppliers.csv	S3	unit_price_eur	75	guesstimate	Assumed/quoted unit price recorded for MemEU (representative Polish memory supplier): EUR 75	Positive EUR/unit value; supplier prices are compared on the same per-component basis in
suppliers.csv	S3	promised_lead_da	5	guesstimate	Model estimate of 5 promised lead days for MemEU (representative Polish memory supplier);	Promised lead time (5 days) is not greater than the modeled mean lead time (7 days).
suppliers.csv	S3	lead_time_mean_days	7	guesstimate	Model estimate for MemEU (representative Polish memory supplier): 7 days total supplier-to-plant assumption; the Polish truck lane L4-TRUCK is 3	Cross-check against network_lanes.csv: route transit component is 3 days; remaining allowance is 4 days. Total matches suppliers.csv.
suppliers.csv	S3	lead_time_sd_day	1	guesstimate	Model assumption: standard deviation of supplier lead time = 1 days for MemEU (representative	Standard deviation is non-negative and smaller than the corresponding mean lead time; it is a
suppliers.csv	S3	capacity_per_month	30000	guesstimate	Model assumption: 30,000 memory components/month available from MemEU (representative Polish memory supplier); no	Capacity is greater than the recorded minimum order quantity (5,000/month) and is expressed in units/month.
suppliers.csv	S3	min_per_month	5000	guesstimate	Model assumption: minimum monthly order quantity of 5,000 memory components from	Minimum is positive and does not exceed the modeled monthly capacity (30,000 units/month).
suppliers.csv	S3	kg_co2e_per_unit	0.4	guesstimate	Model estimate: 0.4 kg CO2e per memory component supplied by MemEU (representative Polish memory supplier); supplier-specific verified	Emission factor is non-negative and expressed per component; used only for the memory-chip supplier-emissions part of the model.
locations.csv	LOC-PLAN	lat	51.05	public	Coordinates for the representative LOC-PLANT location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;

locations.csv	LOC-PLAN	lon	10.43	public	Coordinates for the representative LOC-PLANT location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
locations.csv	LOC-S1	lat	37.57	public	Coordinates for the representative LOC-S1 location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
locations.csv	LOC-S1	lon	126.98	public	Coordinates for the representative LOC-S1 location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
locations.csv	LOC-S2	lat	25.03	public	Coordinates for the representative LOC-S2 location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
locations.csv	LOC-S2	lon	121.57	public	Coordinates for the representative LOC-S2 location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
locations.csv	LOC-S3	lat	52.23	public	Coordinates for the representative LOC-S3 location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
locations.csv	LOC-S3	lon	21.01	public	Coordinates for the representative LOC-S3 location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
locations.csv	LOC-PORT	lat	51.92	public	Coordinates for the representative LOC-PORT location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
locations.csv	LOC-PORT	lon	4.48	public	Coordinates for the representative LOC-PORT location, taken from the geographic reference	Latitude is within −90 to +90 degrees and longitude within −180 to +180 degrees;
network_lanes	L1-SEA	distance_km	19000	guesstimate	Approximate modeled route distance of 19000 km for lane L1-SEA; route is a simplified network	Distance is positive and expressed in km; it is a planning-level route approximation, not a carrier
network_lanes	L1-SEA	transit_days_mea	30	guesstimate	Model estimate of mean transit time = 30 days for lane L1-SEA; used in the inbound supply lead-time	Transit time is positive and expressed in days; lane values are checked against the supplier lead-time
network_lanes	L1-SEA	transit_days_sd	4	guesstimate	Model assumption: transit-time standard deviation = 4 days for lane L1-SEA, representing	Standard deviation is non-negative and smaller than the corresponding mean transit time.
network_lanes	L2-SEA	distance_km	19003	guesstimate	Approximate modeled route distance of 19003 km for lane L2-SEA; route is a simplified network	Distance is positive and expressed in km; it is a planning-level route approximation, not a carrier
network_lanes	L2-SEA	transit_days_mea	43	guesstimate	Model estimate of mean transit time = 43 days for lane L2-SEA; used in the inbound supply lead-time	Transit time is positive and expressed in days; lane values are checked against the supplier lead-time
network_lanes	L2-SEA	transit_days_sd	5	guesstimate	Model assumption: transit-time standard deviation = 5 days for lane L2-SEA, representing	Standard deviation is non-negative and smaller than the corresponding mean transit time.
network_lanes	L3-TRUC	distance_km	500	guesstimate	Approximate modeled route distance of 500 km for lane L3-TRUCK; route is a simplified network	Distance is positive and expressed in km; it is a planning-level route approximation, not a carrier
network_lanes	L3-TRUC	transit_days_mea	2	guesstimate	Model estimate of mean transit time = 2 days for lane L3-TRUCK; used in the inbound supply lead-	Transit time is positive and expressed in days; lane values are checked against the supplier lead-time
network_lanes	L3-TRUC	transit_days_sd	1	guesstimate	Model assumption: transit-time standard deviation = 1 days for lane L3-TRUCK,	Standard deviation is non-negative and smaller than the corresponding mean transit time.
network_lanes	L4-TRUC	distance_km	1100	guesstimate	Approximate modeled route distance of 1100 km for lane L4-TRUCK; route is a simplified network	Distance is positive and expressed in km; it is a planning-level route approximation, not a carrier
network_lanes	L4-TRUC	transit_days_mea	3	guesstimate	Model estimate of mean transit time = 3 days for lane L4-TRUCK; used in the inbound supply lead-	Transit time is positive and expressed in days; lane values are checked against the supplier lead-time
network_lanes	L4-TRUC	transit_days_sd	1	guesstimate	Model assumption: transit-time standard deviation = 1 days for lane L4-TRUCK,	Standard deviation is non-negative and smaller than the corresponding mean transit time.
freight_rates.cs	L1-SEA	rate_eur_per_shi	1500	guesstimate	Model estimate: EUR 1500 charged per shipment on lane L1-SEA; no carrier quotation was supplied.	Rate is positive and expressed per shipment; it is one of the freight-cost inputs.
freight_rates.cs	L1-SEA	rate_eur_per_tk	0.08	guesstimate	Model estimate: EUR 0.08 per tonne-kilometre for lane L1-SEA; used as a distance- and weight-based	Rate is positive and expressed in EUR/tkm; unit is consistent with the freight-cost calculation.
freight_rates.cs	L1-SEA	min_charge_eur	2000	guesstimate	Model estimate: minimum freight charge of EUR 2000 on lane L1-SEA; applied as a floor where the	Minimum charge is non-negative and does not exceed the corresponding shipment-rate
freight_rates.cs	L2-SEA	rate_eur_per_shi	1500	guesstimate	Model estimate: EUR 1500 charged per shipment on lane L2-SEA; no carrier quotation was supplied.	Rate is positive and expressed per shipment; it is one of the freight-cost inputs.
freight_rates.cs	L2-SEA	rate_eur_per_tk	0.08	guesstimate	Model estimate: EUR 0.08 per tonne-kilometre for lane L2-SEA; used as a distance- and weight-based	Rate is positive and expressed in EUR/tkm; unit is consistent with the freight-cost calculation.
freight_rates.cs	L2-SEA	min_charge_eur	2000	guesstimate	Model estimate: minimum freight charge of EUR 2000 on lane L2-SEA; applied as a floor where the	Minimum charge is non-negative and does not exceed the corresponding shipment-rate
freight_rates.cs	L3-TRUC	rate_eur_per_shi	300	guesstimate	Model estimate: EUR 300 charged per shipment on lane L3-TRUCK; no carrier quotation was	Rate is positive and expressed per shipment; it is one of the freight-cost inputs.
freight_rates.cs	L3-TRUC	rate_eur_per_tk	0.35	guesstimate	Model estimate: EUR 0.35 per tonne-kilometre for lane L3-TRUCK; used as a distance- and weight-	Rate is positive and expressed in EUR/tkm; unit is consistent with the freight-cost calculation.
freight_rates.cs	L3-TRUC	min_charge_eur	400	guesstimate	Model estimate: minimum freight charge of EUR 400 on lane L3-TRUCK; applied as a floor where	Minimum charge is non-negative and does not exceed the corresponding shipment-rate
freight_rates.cs	L4-TRUC	rate_eur_per_shi	300	guesstimate	Model estimate: EUR 300 charged per shipment on lane L4-TRUCK; no carrier quotation was	Rate is positive and expressed per shipment; it is one of the freight-cost inputs.
freight_rates.cs	L4-TRUC	rate_eur_per_tk	0.35	guesstimate	Model estimate: EUR 0.35 per tonne-kilometre for lane L4-TRUCK; used as a distance- and weight-	Rate is positive and expressed in EUR/tkm; unit is consistent with the freight-cost calculation.
freight_rates.cs	L4-TRUC	min_charge_eur	400	guesstimate	Model estimate: minimum freight charge of EUR 400 on lane L4-TRUCK; applied as a floor where	Minimum charge is non-negative and does not exceed the corresponding shipment-rate
emission_factors.csv	grid_electricity_DE_2	value	0.344	public	Public German electricity-grid emission factor recorded for 2025: 0.344 kg CO2e/kWh; source URL is retained in the source record.	Factor is positive and expressed per kWh; multiplying by modeled factory electricity use yields the factory-electricity emissions component.

emissio n_facto rs.csv	prima ry_tit aniu m_gl	value	15	guesstima te	Model estimate for primary titanium production: 15 kg CO2e per kg titanium; used as an indicative material factor, not a verified supplier-specific footprint.	Factor is positive and expressed per kg material; applicability depends on titanium mass and the chosen system boundary.
bom.cs v	mem	product_ id	PH-FLAG	guesstima te	Project design choice: PH-FLAG is the product identifier linking this component to the simulated	Matches products.csv product_id; all BOM component rows refer to the same modeled
bom.cs v	mem	compone nt_id	mem	guesstima te	Internal component key “mem” created for this project BOM; not an externally standardized part	Unique internal key within the BOM; matches the component_id value in bom.csv.
bom.cs v	mem	compone nt_name	memory module (12GB+25 6GB)	guesstima te	Representative component description “memory module (12GB+256GB)” for the modeled phone configuration; engineering-model assumption.	Description is non-empty and identifies the modeled component category; not a verified manufacturer part specification.
bom.cs v	mem	qty_per_ unit	1	guesstima te	Model assumption: 1 unit(s) of component mem required per finished phone.	Quantity is positive; component consumption is calculated as quantity per unit multiplied by
bom.cs v	mem	mass_kg	0.005	guesstima te	Model estimate: 0.005 kg per component mem; not a supplier-certified component mass.	Mass is non-negative and expressed in kg per component; BOM component masses can be
bom.cs v	mem	unit_cost _eur	70	guesstima te	Model estimate: EUR 70 per component mem; not a supplier quotation unless separately	Cost is non-negative and expressed in EUR/component; quantity × unit cost contributes
bom.cs v	mem	supplier_ id	S1	guesstima te	Sourcing-model assignment: component mem is assigned to supplier key S1; this is a project	Supplier key is recorded consistently in the BOM; check against suppliers.csv before treating it as an
bom.cs v	mem	added_at _workce	WC1	guesstima te	Process-design assumption: component mem is added at workcenter WC1 according to the	Workcenter reference should exist in workcenters.csv; assignment indicates the
bom.cs v	soc	product_ id	PH-FLAG	guesstima te	Project design choice: PH-FLAG is the product identifier linking this component to the simulated	Matches products.csv product_id; all BOM component rows refer to the same modeled
bom.cs v	soc	compone nt_id	soc	guesstima te	Internal component key “soc” created for this project BOM; not an externally standardized part	Unique internal key within the BOM; matches the component_id value in bom.csv.
bom.cs v	soc	compone nt_name	flagship applicatio n	guesstima te	Representative component description “flagship application processor” for the modeled phone configuration; engineering-model assumption.	Description is non-empty and identifies the modeled component category; not a verified manufacturer part specification.
bom.cs v	soc	qty_per_ unit	1	guesstima te	Model assumption: 1 unit(s) of component soc required per finished phone.	Quantity is positive; component consumption is calculated as quantity per unit multiplied by
bom.cs v	soc	mass_kg	0.003	guesstima te	Model estimate: 0.003 kg per component soc; not a supplier-certified component mass.	Mass is non-negative and expressed in kg per component; BOM component masses can be
bom.cs v	soc	unit_cost _eur	110	guesstima te	Model estimate: EUR 110 per component soc; not a supplier quotation unless separately	Cost is non-negative and expressed in EUR/component; quantity × unit cost contributes
bom.cs v	soc	supplier_ id	S-SOC	guesstima te	Sourcing-model assignment: component soc is assigned to supplier key S-SOC; this is a project	Supplier key is recorded consistently in the BOM; check against suppliers.csv before treating it as an
bom.cs v	soc	added_at _workce	WC1	guesstima te	Process-design assumption: component soc is added at workcenter WC1 according to the	Workcenter reference should exist in workcenters.csv; assignment indicates the
bom.cs v	pcb	product_ id	PH-FLAG	guesstima te	Project design choice: PH-FLAG is the product identifier linking this component to the simulated	Matches products.csv product_id; all BOM component rows refer to the same modeled
bom.cs v	pcb	compone nt_id	pcb	guesstima te	Internal component key “pcb” created for this project BOM; not an externally standardized part	Unique internal key within the BOM; matches the component_id value in bom.csv.
bom.cs v	pcb	compone nt_name	printed circuit board	guesstima te	Representative component description “printed circuit board” for the modeled phone configuration; engineering-model assumption.	Description is non-empty and identifies the modeled component category; not a verified manufacturer part specification.
bom.cs v	pcb	qty_per_ unit	1	guesstima te	Model assumption: 1 unit(s) of component pcb required per finished phone.	Quantity is positive; component consumption is calculated as quantity per unit multiplied by
bom.cs v	pcb	mass_kg	0.02	guesstima te	Model estimate: 0.02 kg per component pcb; not a supplier-certified component mass.	Mass is non-negative and expressed in kg per component; BOM component masses can be
bom.cs v	pcb	unit_cost _eur	15	guesstima te	Model estimate: EUR 15 per component pcb; not a supplier quotation unless separately	Cost is non-negative and expressed in EUR/component; quantity × unit cost contributes
bom.cs v	pcb	supplier_ id	S-PCB	guesstima te	Sourcing-model assignment: component pcb is assigned to supplier key S-PCB; this is a project	Supplier key is recorded consistently in the BOM; check against suppliers.csv before treating it as an
bom.cs v	pcb	added_at _workce	WC1	guesstima te	Process-design assumption: component pcb is added at workcenter WC1 according to the	Workcenter reference should exist in workcenters.csv; assignment indicates the
bom.cs v	disp	product_ id	PH-FLAG	guesstima te	Project design choice: PH-FLAG is the product identifier linking this component to the simulated	Matches products.csv product_id; all BOM component rows refer to the same modeled
bom.cs v	disp	compone nt_id	disp	guesstima te	Internal component key “disp” created for this project BOM; not an externally standardized part	Unique internal key within the BOM; matches the component_id value in bom.csv.
bom.cs v	disp	compone nt_name	6.7-inch LTPO OLED	guesstima te	Representative component description “6.7-inch LTPO OLED display” for the modeled phone configuration; engineering-model assumption.	Description is non-empty and identifies the modeled component category; not a verified manufacturer part specification.
bom.cs v	disp	qty_per_ unit	1	guesstima te	Model assumption: 1 unit(s) of component disp required per finished phone.	Quantity is positive; component consumption is calculated as quantity per unit multiplied by
bom.cs v	disp	mass_kg	0.06	guesstima te	Model estimate: 0.06 kg per component disp; not a supplier-certified component mass.	Mass is non-negative and expressed in kg per component; BOM component masses can be
bom.cs v	disp	unit_cost _eur	90	guesstima te	Model estimate: EUR 90 per component disp; not a supplier quotation unless separately	Cost is non-negative and expressed in EUR/component; quantity × unit cost contributes
bom.cs v	disp	supplier_ id	S-DISP	guesstima te	Sourcing-model assignment: component disp is assigned to supplier key S-DISP; this is a project	Supplier key is recorded consistently in the BOM; check against suppliers.csv before treating it as an

bom.csv	disp	added_at_workce	WC3	guesstimate	Process-design assumption: component disp is added at workcenter WC3 according to the	Workcenter reference should exist in workcenters.csv; assignment indicates the
bom.csv	batt	product_id	PH-FLAG	guesstimate	Project design choice: PH-FLAG is the product identifier linking this component to the simulated	Matches products.csv product_id; all BOM component rows refer to the same modeled
bom.csv	batt	component_id	batt	guesstimate	Internal component key “batt” created for this project BOM; not an externally standardized part	Unique internal key within the BOM; matches the component_id value in bom.csv.
bom.csv	batt	component_name	5000 mAh	guesstimate	Representative component description “5000 mAh battery” for the modeled phone	Description is non-empty and identifies the modeled component category; not a verified
bom.csv	batt	qty_per_unit	1	guesstimate	Model assumption: 1 unit(s) of component batt required per finished phone.	Quantity is positive; component consumption is calculated as quantity per unit multiplied by
bom.csv	batt	mass_kg	0.05	guesstimate	Model estimate: 0.05 kg per component batt; not a supplier-certified component mass.	Mass is non-negative and expressed in kg per component; BOM component masses can be
bom.csv	batt	unit_cost_eur	18	guesstimate	Model estimate: EUR 18 per component batt; not a supplier quotation unless separately	Cost is non-negative and expressed in EUR/component; quantity × unit cost contributes
bom.csv	batt	supplier_id	S-BATT	guesstimate	Sourcing-model assignment: component batt is assigned to supplier key S-BATT; this is a project	Supplier key is recorded consistently in the BOM; check against suppliers.csv before treating it as an
bom.csv	batt	added_at_workce	WC3	guesstimate	Process-design assumption: component batt is added at workcenter WC3 according to the	Workcenter reference should exist in workcenters.csv; assignment indicates the
bom.csv	cam	product_id	PH-FLAG	guesstimate	Project design choice: PH-FLAG is the product identifier linking this component to the simulated	Matches products.csv product_id; all BOM component rows refer to the same modeled
bom.csv	cam	component_id	cam	guesstimate	Internal component key “cam” created for this project BOM; not an externally standardized part	Unique internal key within the BOM; matches the component_id value in bom.csv.
bom.csv	cam	component_name	triple camera module	guesstimate	Representative component description “triple camera module (OIS)” for the modeled phone configuration; engineering-model assumption.	Description is non-empty and identifies the modeled component category; not a verified manufacturer part specification.
bom.csv	cam	qty_per_unit	1	guesstimate	Model assumption: 1 unit(s) of component cam required per finished phone.	Quantity is positive; component consumption is calculated as quantity per unit multiplied by
bom.csv	cam	mass_kg	0.015	guesstimate	Model estimate: 0.015 kg per component cam; not a supplier-certified component mass.	Mass is non-negative and expressed in kg per component; BOM component masses can be
bom.csv	cam	unit_cost_eur	55	guesstimate	Model estimate: EUR 55 per component cam; not a supplier quotation unless separately	Cost is non-negative and expressed in EUR/component; quantity × unit cost contributes
bom.csv	cam	supplier_id	S-CAM	guesstimate	Sourcing-model assignment: component cam is assigned to supplier key S-CAM; this is a project	Supplier key is recorded consistently in the BOM; check against suppliers.csv before treating it as an
bom.csv	cam	added_at_workce	WC3	guesstimate	Process-design assumption: component cam is added at workcenter WC3 according to the	Workcenter reference should exist in workcenters.csv; assignment indicates the
bom.csv	chss	product_id	PH-FLAG	guesstimate	Project design choice: PH-FLAG is the product identifier linking this component to the simulated	Matches products.csv product_id; all BOM component rows refer to the same modeled
bom.csv	chss	component_id	chss	guesstimate	Internal component key “chss” created for this project BOM; not an externally standardized part	Unique internal key within the BOM; matches the component_id value in bom.csv.
bom.csv	chss	component_name	titanium chassis	guesstimate	Representative component description “titanium chassis” for the modeled phone configuration;	Description is non-empty and identifies the modeled component category; not a verified
bom.csv	chss	qty_per_unit	1	guesstimate	Model assumption: 1 unit(s) of component chss required per finished phone.	Quantity is positive; component consumption is calculated as quantity per unit multiplied by
bom.csv	chss	mass_kg	0.05	guesstimate	Model estimate: 0.05 kg per component chss; not a supplier-certified component mass.	Mass is non-negative and expressed in kg per component; BOM component masses can be
bom.csv	chss	unit_cost_eur	20	guesstimate	Model estimate: EUR 20 per component chss; not a supplier quotation unless separately	Cost is non-negative and expressed in EUR/component; quantity × unit cost contributes
bom.csv	chss	supplier_id	S-CHSS	guesstimate	Sourcing-model assignment: component chss is assigned to supplier key S-CHSS; this is a project	Supplier key is recorded consistently in the BOM; check against suppliers.csv before treating it as an
bom.csv	chss	added_at_workce	WC3	guesstimate	Process-design assumption: component chss is added at workcenter WC3 according to the	Workcenter reference should exist in workcenters.csv; assignment indicates the
bom.csv	oth	product_id	PH-FLAG	guesstimate	Project design choice: PH-FLAG is the product identifier linking this component to the simulated	Matches products.csv product_id; all BOM component rows refer to the same modeled
bom.csv	oth	component_id	oth	guesstimate	Internal component key “oth” created for this project BOM; not an externally standardized part	Unique internal key within the BOM; matches the component_id value in bom.csv.
bom.csv	oth	component_name	cables/screws/haptics/pack	guesstimate	Representative component description “cables/screws/haptics/packaging” for the modeled phone configuration; engineering-model	Description is non-empty and identifies the modeled component category; not a verified manufacturer part specification.
bom.csv	oth	qty_per_unit	1	guesstimate	Model assumption: 1 unit(s) of component oth required per finished phone.	Quantity is positive; component consumption is calculated as quantity per unit multiplied by
bom.csv	oth	mass_kg	0.02	guesstimate	Model estimate: 0.02 kg per component oth; not a supplier-certified component mass.	Mass is non-negative and expressed in kg per component; BOM component masses can be
bom.csv	oth	unit_cost_eur	12	guesstimate	Model estimate: EUR 12 per component oth; not a supplier quotation unless separately	Cost is non-negative and expressed in EUR/component; quantity × unit cost contributes
bom.csv	oth	supplier_id	S-OTH	guesstimate	Sourcing-model assignment: component oth is assigned to supplier key S-OTH; this is a project	Supplier key is recorded consistently in the BOM; check against suppliers.csv before treating it as an
bom.csv	oth	added_at_workce	WC5	guesstimate	Process-design assumption: component oth is added at workcenter WC5 according to the	Workcenter reference should exist in workcenters.csv; assignment indicates the

Printed output of the CheckCell.

```
PASS  T1 Format
      all 10 files present, columns exact, no blanks or negatives
PASS  T2 Capacity
      WC1 SMT: busy 64.8% of available time
      WC2 Board test: busy 32.4% of available time
      WC3 Assembly: busy 51.5% of available time
      WC4 Final test: busy 48.6% of available time
      WC5 Packaging: busy 15.4% of available time
PASS  T3 Little's Law
      run 1: L = 2.84  vs  λxW = 17.499/h x 0.16 h = 2.84  (gap 0.0%)
      10 run(s) checked
PASS  T4 Flow balance
      run 1: started 36,400, finished 36,398, key components used 36,400 (allowed 36,398 to 36,400)
PASS  T5 Margin
      cost per unit 401.42 to 401.48 EUR vs list price 755.46 EUR
--------------------------------------------------------------------------------
5 of 5 tests passed  ·  ALL PASS
check cell v1.0 · cell 3ad30383c119 · data de550de722d3 · 2026-11-10 19:35
```

Key Arguments of the papers:

1. **Choosing between single and multiple sourcing based on supplier default risk: A real options approach**

The paper argues that relying on a single supplier can expose a company to significant supply risks if that supplier experiences a disruption or failure. Using a real-options approach and Monte Carlo simulation, the authors show how multiple sourcing can provide flexibility by allowing orders to be shifted to another existing supplier when the main supplier cannot deliver. However, this flexibility comes with additional costs, so companies must weigh the potential benefits of reduced supply risk against the costs of managing multiple suppliers.

Costantino, N., & Pellegrino, R. (2010). Choosing between single and multiple sourcing based on supplier default risk: A real options approach. Journal of Purchasing and Supply Management, 16(1), 27–40. https://doi.org/10.1016/j.pursup.2009.08.001

2. **Budget-constrained dual sourcing with option contracts under disruption risk.**

The paper examines how companies can use dual sourcing when they face both supply disruption risks and limited procurement budgets. It analyses a strategy involving a reliable but more expensive supplier and a less reliable, lower-cost supplier, using option contracts to provide flexibility when disruptions occur. The findings show that budget availability, disruption probability and demand uncertainty influence whether single or dual sourcing is the more suitable strategy.

Wang, X., Tang, J., Chen, B., & Hou, B. (2026). *Budget-constrained dual sourcing with option contracts under disruption risk.* Omega. https://doi.org/10.1016/j.omega.2026.103682

AI Note for the project:

1. How we used AI

- Used ChatGPT (ChatGPT Go) to understand and analyse CSV files.
- Used it to develop and improve the Python simulation for the three scenarios: Baseline, Red Sea Crisis and Dual Sourcing.
- Used it to calculate production output, costs and CO_2 emissions and improve the simulation visualisation.
- Asked ChatGPT to create a detailed prompt for Gemini to design the poster based on our reference images.
- Used ChatGPT again to review the poster and compare its figures with our simulation results.

2. How we checked the AI output

- Compared the generated code and results with our original CSV files and simulation outputs.
- Manually reviewed calculations and checked whether the figures matched our data.
- Treated modelling assumptions as estimates rather than verified historical facts.
- Corrected information wherever we found inconsistencies.

3. Mistakes we identified

- The initial CO_2 calculation was incomplete because some supplier and transport emissions were not fully considered.
- Some figures in the AI-generated poster did not match our actual simulation results.
- We identified these issues by comparing the output with our source data and simulation results, and manually checked the figures before using them.